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IoT-Enabled Smart Car Parking System through Integrated Sensors and Mobile Application

Abdullah Al Mamun*, Abdul Hasib†, Abu Salyh Muhammad Mussa‡, Rakib Hossen§, and Anichur Rahman¶
Department of IRE & CySE, Bangabandhu Sheikh Mujibur Rahman Digital University, Bangladesh,
Department of CSE, National Institute of Textile Engineering and Research (NITER),
Constituent Institute of the University of Dhaka, Savar, Dhaka-1350, Bangladesh
shaikat.bdu@gmail.com*, sm.abdulhasib.bd@gmail.com†, abusalyh43@gmail.com ‡, rakib0001@bdu.ac.bd§, and anis_cse@niter.edu.bd¶

***Abstract*—The growing urban population and increasing car ownership have led to a critical demand for efficient parking management solutions. This paper presents an IoT-enabled smart car parking system integrating IR sensors for vehicle detection, DHT22 for temperature and humidity monitoring, and MQ-2 sensors for hazardous gas detection. Key components include servo motors for automated gate control, OLED displays for real-time parking updates, and a mobile app utilizing MQTT protocol for seamless communication. The system offers real-time monitoring, automated slot management, and enhanced safety measures such as exhaust fan activation during high gas concentrations. Experimental results demonstrate improved parking efficiency, reduced emissions, and enhanced user experience, setting a new benchmark for smart parking systems in urban infrastructure.**

***Index Terms*—Internet of Things, Smart Parking System, Arduino UNO, Raspberry Pi, MQTT Protocol, Safety, Real-time Monitoring, Environmental Monitoring, Mobile App Interface.**

I. Introduction

The majority of people on the planet reside in cities. Therefore, cities are now fully occupied. According to our research on transport and environment, passenger car transit increased by 18% from 2000 to 2019 and road freight by 31% [1]. The global smart parking market is expected to be valued at around US$5.1 billion in revenue in 2023. According to IMARC Group, the market is projected to grow to nearly $13.8 billion by the close of 2032, at a CAGR of 11.3% during the forecast period (2024–2032) [2]. Cities are currently responsible for more than 75% of the garbage produced, 80% of CO2 emissions, and 75% of the energy consumed [3]. According to EEA data, CO2 emissions from passenger automobiles in the 27 EU Member States increased by 5.8%, while emissions from heavy freight vehicles climbed by 5.5%, from 2000 to 2019 [4]. According to researchers, traffic congestion caused by persons looking for parking accounts for 30 percent of road transport emissions. Previous research reveals that hunting for parking produces 30% of the congestion in populated places. Drivers spend an average of 7.8 minutes looking for a spot [5]. Moreover, in the parking lots of shopping complexes, many particularly unkind drivers who are agitated by the inadequate availability of parking spaces frequently occupy the parking spaces specifically made available for physically challenged people such as the aged and disabled, hence encouraging anti-social behavior. Smart parking systems can increase the quality of users' lives by simplifying their daily activities, reducing fuel consumption and emissions, and mitigating congestion in cities. These parking systems will help ease traffic congestion by providing information on the availability of parking spaces in public and private facilities, thus giving comfort to users and improving user experience [6], [7].

On the other hand, IoT is a key enabler in realizing the concept of smart cities, where one of its important components is enhancing car parking and traffic facilities [8]. It has become a challenge for drivers in modern-day cities to look for a parking slot due to the excessive amount of private car users. Smart cities are supposed to rationalize their parking assets, which will help minimize searching, congestion, and accidents as well. If drivers are provided with prior knowledge of available parking slots at their desired location and nearby destinations, this will ultimately fix the problems related to parking and traffic congestion [9]. Indeed, smart parking solutions play a critical role in achieving enhanced urban mobility and a sustainable environment.

In this paper, an IoT-enabled smart car parking system through integrated sensors and mobile applications is proposed. This system includes IR sensors for the exact detection of vehicles, DHT22 sensors for temperature and humidity, and MQ2 gas sensors to detect hazardous gases. Core components of Arduino UNO and Raspberry Pi microcontrollers are responsible for gate operations, ventilation, and data communication via MQTT. The mobile app will provide real-time parking availability information to users, improving the overall user experience. In this regard, the main contributions of the paper could be understood as follows:

- In this paper, IR sensors are employed for vehicle detection, DHT22 sensors for temperature and humidity measurement, and MQ2 sensors for hazardous gases to ensure real-time data.
- The authors make use of the Arduino UNO and Raspberry Pi microcontrollers in automating gate operations with exhaust fan management at the center of the Smart Car Parking System to monitor environmental conditions.
- We use the MQTT protocol in this Smart Car Parking System to provide real-time updates through a mobile app, which shows parking availability and environmental conditions on an OLED screen.
- This system demonstrates high accuracy in vehicle detection and environmental monitoring, significantly improving parking management efficiency and user satisfaction with automated, real-time updates.

Overall, the proposed solution combines automation, IoT technology, and real-time data processing for improved operational performance and user convenience. It provides a stable and scalable response to contemporary parking difficulties.

Organization: This paper is structured as follows: Section II provides an in-depth review of related work. Section III presents the methodology, system architecture, and detailed implementation of our IoT-based smart parking solution. Section IV discusses the real-time analysis and experimental results. Further, Section V addresses the limitations encountered during the study and proposes potential future enhancements. Finally, this paper concludes in Section VI.

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TABLE I
COMPARISON WITH THE EXISTING SMART PARKING SYSTEMS

Study	Technologies Used	Key Features	Strengths	Weaknesses
Nikitha et al. (2021) [10]	Raspberry Pi, IR sensors, camera	Vehicle detection, license plate capture	High accuracy	No online booking, limited sensors
Won et al. (2021) [11]	IR sensors, cloud, mobile app	Real-time updates, remote reservation	Reduces search time, fuel	Sensor calibration, internet dependency
Sofian et al. (2022) [12]	ESP32, light sensors (LDR)	Space detection, web app updates	Effective in crowded areas	Complex wiring, sensor sensitivity
Simatupang et al. (2022) [13]	ESP32, ultrasonic sensors	Online slot check, reservation	Enables reservations	Update delay, unauthorized parking
Venu et al. (2022) [14]	RFID, IR Sensors, Node MCU, Firebase, MIT App Inventor	Slot monitoring, remote booking, RFID access, cloud sync	Efficient, secure, remote access, real-time updates	Potential sensor errors, setup costs
Vinayak et al. (2022) [15]	RFID, Raspberry Pi	Space identification, vehicle tracking	Centralized tracking	High setup costs, maintenance
Amran et al. (2023) [16]	IR sensors, ESP32, servo motors	Space availability, gate control	Improved user experience	Technical reliability, user adherence, scalability
Singh et al. (2023) [17]	Arduino, ESP8266, IR sensors	Real-time mapping, slot updates, remote booking	Less intervention, efficient space use, user-friendly, real-time info	Sensor accuracy, connectivity issues, setup costs, RFID misuse
Kusuma et al. (2023) [18]	Arduino, ESP32-CAM, ultrasonic, PIR sensors	Touchless operation, gate control	High success rates	WiFi dependency, sensor angle limitations
Shaikh et al. (2024) [19]	Raspberry Pi, Pi camera, ultrasonic sensor	Real-time detection, data processing, occupancy verification	Reduces search time, real-time updates	Sensor calibration, setup complexity
Proposed system	IR sensors, DHT22, MQ-2, Raspberry Pi, mobile app, servo motors, buzzers, OLED, MQTT	Vehicle detection, environment monitoring, automated control, spot availability indication	Holistic approach, real-time data, seamless integration	Setup complexity, MQTT dependency

II. RELATED WORKS

Various systems were proposed to address smart-parking problems. Nikitha et al. [10] developed a Raspberry Pi-based system using IR sensors and an LCD display without online booking and advanced sensor integration. Won et al. [11] introduced an IoT system implemented using IR sensors, cloud databases, and a mobile app providing real-time updates and remote reservations but with a dependency on sensor calibration and internet stability. Meanwhile, Sofian et al. [12] introduced the ESP32-based system that uses light sensors for real-time parking data on a website and a mobile app, yet faced the problem of overwhelming assembling and sensor sensitivity. Simatupang et al. [13] designed a system with ESP32 and ultrasonic sensors for online reservations, but update delays and wrongful parking were the issues. In the same vein, Venu et al. [14] designed RFID and IoT-based secure entry as well as cloud-based booking, which faced the challenges of sensor failure and low-internet connectivity. Vinayak et al. [15] developed RFID and Raspberry Pi implemented solutions facing high setup costs as well as heavy maintenance problems.

Other works are also notable like Amran et al.'s [16] system that employed infrared sensors, ESP32 microcontrollers, and servo motors for parking space availability, but the system was found not reliable and not scalable. Singh et al. [17] presented SCPS with IR sensors, Arduino, and NodeMCU ESP8266, which offered real-time booking but were difficult to maintain due to accurate sensors required both for booking as well as connectivity. Kusuma et al. [18] developed a touchless system using Arduino and ESP32-CAM, but it suffered from a lack of WiFi stability as well as sensor angles. Shaikh et al. [19] proposed a system using Raspberry Pi, camera, and ultrasonic sensor for reducing parking search time, but again, the system had the disadvantage of WiFi dependency and the need for calibration.

The proposed IoT-enabled intelligent car parking will integrate IR sensors for vehicle detection, DHT22 for temperature and humidity, MQ-2 for air quality, servo motors for gate automation, buzzers for alerts, and an OLED display for real-time slot availability. The mobile app connected through MQTT shall provide real-time monitoring and control, hence ensuring far better integration, automation, and responsiveness than traditional systems devoid of IoT. Table I highlights the differences and similarities between our work and other studies, showing how our system integrates various sensors and communication technologies to enhance the parking system.

III. PROPOSED METHODOLOGY FOR IOT-ENABLED SMART CAR PARKING SYSTEM

In our system, we implement an advanced IoT-enabled solution integrating various sensors and MCUs for advanced parking management. Each slot has an IR sensor for vehicle detection, ensuring accurate entry and exit monitoring. A DHT22 sensor continuously measures temperature and humidity, displaying data on an OLED screen. An MQ2 gas sensor detects hazardous gases, activating an exhaust fan when needed. MQTT protocol facilitates real-time updates on slot availability and environmental conditions via a mobile app [20]. These components' comprehensive workflow and interactions are depicted in Fig. 1, illustrating the block diagram of our IoT-based smart car parking system.

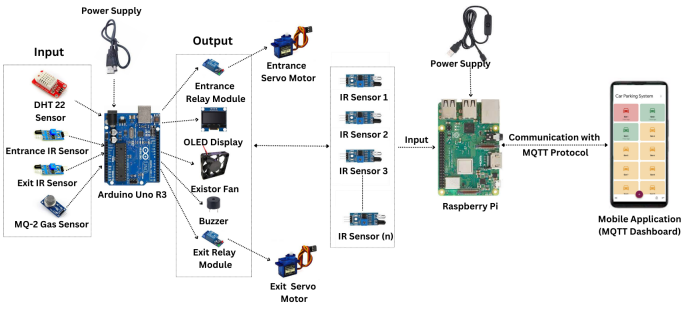


Fig. 1. Proposed Block Diagram for Controlling each Connection

A. Collecting of Hardware Requirements

This system requires several essential hardware components to function effectively. Table II summarizes the specific roles and functionalities of each hardware component within the smart car parking system. The hardware components were chosen for their affordability, availability, and suitability. IR

sensors ensure reliable vehicle detection, DHT22 sensors provide accurate temperature and humidity monitoring, and the MQ-2 gas sensor effectively detects gas levels for safety. Raspberry Pi handles data processing and communication, while Arduino Uno enables efficient sensor interfacing and low-power operation, creating a cost-effective and efficient system.

TABLE II
FUNCTIONALITIES OF HARDWARE COMPONENTS IN THE SYSTEM

Component	Functionality
Arduino Uno	Controls entrance and exit gates with IR sensors. Manages servo motors and integrates MQ2 gas sensors for hazardous gas monitoring. Activates exhaust fan and buzzer for safety alerts.
IR Sensors	Detects vehicles at entrance and exit gates. Monitors occupancy status of parking slots.
Relay Module	Interfaces with Arduino Uno to control servo motors for gate operation.
Servo Motors	Operates entrance and exit gates based on signals from Arduino Uno via a relay module.
MQ2 Gas Sensor	Monitors gas concentrations in the parking area. Triggers exhaust fan via Arduino Uno for ventilation.
Exhaust Fan	Activates based on MQ2 sensor readings for proper ventilation.
Buzzer	Alerts audibly when vehicles approach the entrance gate. Connected to Arduino Uno.
DHT22 Sensor	Measures temperature and humidity inside the parking facility. Provides real-time data to Arduino Uno.
OLED Display	Shows real-time environmental data from the DHT22 sensor. Displays parking slot availability updates.
Raspberry Pi	Collects data from IR sensors at parking slots. Sends real-time updates via MQTT to the mobile app. Enables monitoring and management of parking slots.

B. Arrangement of Software Requirements

1) Arduino IDE: The Arduino IDE supports Linux, Mac OS X, and Windows, facilitating programming in C and C++. Sketches are created and uploaded to Arduino boards via the IDE for execution [21].

2) Raspberry Pi OS: Raspberry Pi OS is a Debian-based Unix-like OS for Raspberry Pi boards (excluding Pico). It features a modified LXDE desktop with Openbox and comes with Wolfram Mathematica, VLC, and a lightweight Chromium browser [22].

3) Putty and Advanced IP scanner: To connect and upload code to a Raspberry Pi (RPi), we use tools like PuTTY and Advanced IP Scanner. We open PuTTY, enter the IP to SSH in, and log in. Navigate to our code directory using terminal commands (cd) and use PuTTY's SCP feature to transfer files from our local machine to the RPi [23].

4) Mobile Application: MQTT Dashboard is an Android application designed for easy interaction, control, and management of MQTT-enabled devices [24].

C. Analysis of System Design

Our IoT-enabled Parking System efficiently manages space availability, monitors environmental conditions, and ensures proper ventilation, enhancing user experience and maintaining optimal conditions. It uses a Raspberry Pi for MQTT communication and an Arduino Uno for controls and sensor integrations. In Algorithm 1, The pseudocode represents the logic for the smart car parking system. It outlines how the system manages parking slots, updates displays, and communicates status through MQTT.

When activated, the system begins monitoring all sensors and controls. As a car approaches, the entrance IR sensor detects the vehicle (V_{in}). If no car is detected ($V_{\text{in}} = 0$), the entrance gate remains closed. If a car is detected ($V_{\text{in}} = 1$), the system checks for available parking spaces. If no space is available (Total vacant $= 0$), the entrance gate remains closed. If a space is available (Total vacant > 0), the entrance gate opens via a relay and servo motor, and a buzzer activates ($B = 1$). The system continuously monitors temperature (T) and humidity (H) using a DHT22 sensor. This information, along with the parking slots count, is displayed on an OLED screen (Equation 1):

Algorithm 1 IoT-Enabled Smart Car Parking System

```
Init(S, T, H, G, F, D)                                  ▷ Initialize components
S = [0, 0, ..., 0]                                      ▷ Set all parking slots to vacant
while Active do
    if IR_in == 1 then                                  ▷ Car detected at entrance
        Total_vacant− = 1                               ▷ Decrease vacant slots
        if Total_vacant ≥ 0 then                        ▷ Slots available
            G_in = 1                                    ▷ Open entrance gate
            B = 1                                       ▷ Turn on buzzer
            D(T, H, Total_vacant, Total_slots)          ▷ Update Display
            MQTT(T_P, S)                                ▷ Publish status to MQTT
        else                                            ▷ No slots available
            Total_vacant+ = 1                           ▷ Revert vacant slots
        end if
    end if
    if IR_out == 1 then                                 ▷ Car detected at exit
        G_out = 1                                       ▷ Open exit gate
        Total_vacant+ = 1                               ▷ Increase vacant slots
        D(T, H, Total_vacant, Total_slots)              ▷ Update Display
        MQTT(T_P, S)                                    ▷ Publish status to MQTT
    end if
    for i = 1 to n do                                   ▷ Update slot status
        S_i = IR_slot_i                                 ▷ Read slot status
        MQTT(T_P_i, S_i)                                ▷ Publish slot status
    end for
    T, H = DHT22                                        ▷ Get temperature and humidity
    D(T, H, Total_vacant, Total_slots)
    G = MQ2                                             ▷ Read gas level
    if G > G_th then                                    ▷ Gas threshold exceeded
        F = 1                                           ▷ Activate exhaust fan
    end if
end while
Deinit(S, T, H, G, F, D)                                ▷ Shutdown components
```

$$\text{Display} = f(T, H, \text{Total vacant}, \text{Total slots}) \tag{1}$$

When a car exits, the exit IR sensor detects the vehicle (V_{out}). If no car is detected ($V_{\text{out}} = 0$), the exit gate remains closed. If a car is detected ($V_{\text{out}} = 1$), the exit gate opens via a relay and servo motor. The available slots count is then updated.

The system integrates with a mobile app "MQTT Dashboard" via MQTT communication. Clients connect to the app using the provided IP address and port ($\text{IP}_{\text{app}}, \text{port}_{\text{app}}$). The real-time status of parking slots is communicated as follows:

The MQTT communication between the Raspberry Pi and the mobile app involves publishing and subscribing to topics. Let:

- S be the state of the parking slot ($S = 1$ for occupied, $S = 0$ for vacant).
- T_P be the topic for parking slot status updates.
- D_R be the data rate, delay be the communication latency, and EC be the error correction rate.

The parking slot status is updated and published to the topic T_P (Equation 2):

$$\text{Publish}(T_P, S) \implies \text{Message}(S) \tag{2}$$

Clients subscribe to the topic to receive updates (Equation 3):

$$\text{Subscribe}(T_P) \implies \text{Receive Message}(S) \tag{3}$$

The data transfer rate (D_R) for MQTT communication is given by (Equation 4):

$$D_R = \frac{\text{Data transmitted}}{\text{Time}} \tag{4}$$

The communication delay (delay) is calculated as (Equation 5):

$$\text{delay} = \text{time}_{\text{end}} - \text{time}_{\text{start}} \tag{5}$$

The error correction rate (EC) is given by (Equation 6):

$$EC = \frac{\text{Errors corrected}}{\text{Total errors}} \tag{6}$$

Upon entering the parking lot, clients can view available slots on their mobile devices and park in the nearest and easiest vacant slot. The parking slot status is updated through MQTT communication, changing from green (available) to red (occupied) on the mobile app. Additionally, the system includes an MQ2 gas sensor to monitor the parking lot for hazardous gas levels (G). If the gas concentration exceeds a predefined threshold ($G > G_{\text{threshold}}$), an exhaust fan activates ($F = 1$) to ventilate the area (Equation 7):

$$G > G_{\text{threshold}} \implies F = 1 \tag{7}$$

The probability of the parking lot being fully occupied (P_{full}) can be modeled using a Poisson distribution with arrival rate λ. The probability that the number of cars k equals the total slots n is (Equation 8):

$$P_{\text{full}} = \frac{\lambda^n e^{-\lambda}}{n!} \tag{8}$$

If the average time a car spends in the lot is T_{avg}, and the arrival rate is λ, then the average number of cars in the lot (L) can be estimated using Little's Law (Equation 9):

$$L = \lambda \cdot T_{\text{avg}} \tag{9}$$

The response time of the ventilation system (t_{response}) when gas concentration exceeds the threshold is (Equation 10):

$$t_{\text{response}} = \frac{\Delta G}{r} \tag{10}$$

Where ΔG is the difference between the current gas concentration and the threshold, and r is the rate of gas concentration reduction. When turned off, all monitoring and control operations cease, concluding the workflow.

D. Working Procedure of System Model

Our IoT-enabled Smart Car Parking System is designed to efficiently manage parking availability, monitor environmental conditions, and provide real-time updates to users through a mobile application. The flowchart in Fig. 2 illustrates the whole System. This kind of system allows data handling to be more flexible and real-time and enhances user experience in maintaining ideal conditions within the parking lot as well.

When turned on, the system reads temperature and humidity from a DHT22 sensor, showing the data and parking availability on an OLED screen. An entrance IR sensor acts as an entrance slot for incoming vehicles, and the gate will open if the space is available through the servo/relay motor while a buzzer sounds. The slot count changes automatically. The status of the parking slots can be viewed online through MQTT communication using IP and port connections via the "MQTT Dashboard" mobile application. In this application, free slots are highlighted in green and occupied slots in red. These come in real time. As soon as a vehicle exits, the IR sensor will activate the gate to open and decrease the number count, updating the app status from red to green. Additionally, an MQ-2 gas sensor monitors hazardous gas levels. If concentrations exceed safe thresholds, an exhaust fan activates to ensure safety. The system can be turned off, ceasing all operations and completing its workflow. This automated design enhances efficiency, safety, and user satisfaction in parking management.

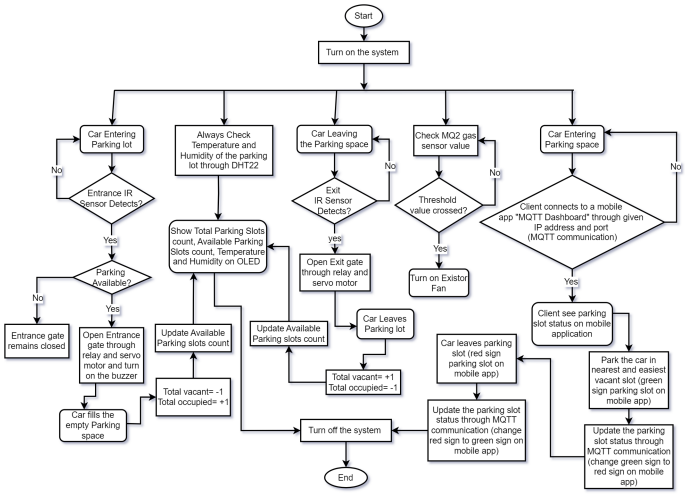


Fig. 2. Workflow Diagram of Proposed System

IV. Result Analysis and Performance Measurement

A. Performance Analysis

1) Performance Analysis of IR Sensor: Fig. 3(a) analyzes how an IR sensor can detect vehicle entry and exit with different light intensities for a whole week. The sensor's accuracy reduced by a whopping 52% to 98% of what it initially was as the ambient light intensity got higher and, therefore, the calibration and screening became necessary, which is why the sensor's performance exhibited. For optimal parking management, it is important to provide high detection accuracy, which will reduce false detection. IR sensors are enclosed in protective casings to prevent dust intrusion, with anti-glare coatings to reduce light interference.

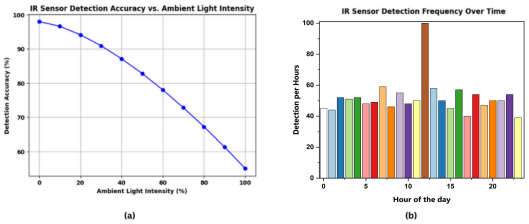


Fig. 3. (a) IR Sensor Detection Accuracy vs. Ambient Light Intensity, (b) IR Sensor Detection Frequency

Fig. 3(b) shows the frequency of IR sensor detection in the CMH parking area, Dhaka Cantonment, Bangladeshover 24 hours. The peak detection is up to 100 per hour between 12 PM and 1 PM. It is used to help manage the flow of traffic, allocate resources, maintain system efficiency, etc. Knowing peak hours helps to optimize the schedules and allow for dynamic pricing. The system's modular design ensures scalability for larger parking facilities by deploying multiple NodeMCU and Raspberry Pi units via MQTT. The MQTT dashboard application supports more than 100 slots simultaneously, making it suitable for managing and monitoring large parking areas efficiently. However, limitations such as increased latency in data transmission and higher network congestion may arise. These can be addressed by integrating edge computing techniques and upgrading to more

robust hardware, like Raspberry Pi 4 with higher processing capabilities.

2) Performance Analysis of DHT22 Temperature and Humidity Sensor: Fig. 4(a) shows the temperature and humidity variation in the parking area for four days with temperature ranging from 24°C to 35°C, Humidity was varied between 63% and 85%. The entrance and exit of vehicles, each step also introduces temperature humidity until the degree of comfort as well as safety used in car parking areas.

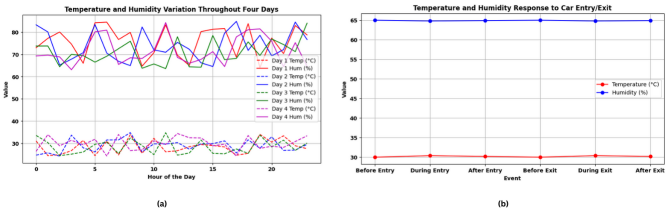


Fig. 4. (a) Temperature and Humidity Value Monitoring, (b) Temperature and Humidity Response to Car Entry/Exit

On the other hand, Fig. 4(b) illustrates temperature and humidity measurements before car entry and exit distinguished into the different states. Upon entry, the temperature rose by 0.4°C and the humidity by 0.2%, with a slight decrease of both upon leaving. By getting insights into those alterations, appropriate conditions for the facilities can be established.

3) Performance Analysis of MQ-2 Gas Sensor: Fig. 5(a) shows the gas concentration levels (e.g., CO) in the parking area for four days ranging from 1 ppm to 25 ppm at peak hours. This information is very critical to maintain air quality and safety.

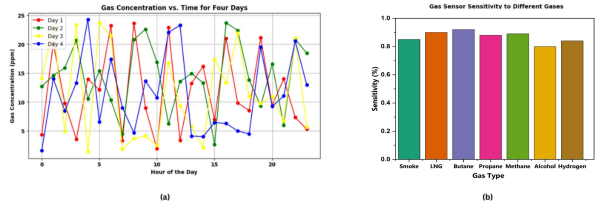


Fig. 5. (a) Gas Value Monitoring, (b) Comparison of Different Gas Sensor's Sensitivity

Fig. 5(b) is a graph on the sensitivity of the MQ-2 sensor in relation to various gases, where the highest sensitivity is to Butane — 0.92 and the lowest to Alcohol — 0.80. This ensures comprehensive air quality monitoring. The MQ-2 gas sensor is initially calibrated in the system setup by using standard calibration gases. For frequent, shorter monitoring cycles, after three months, or depending on the environment, recalibration is suggested. The life of the sensor is estimated to be about two years, depending on the frequency of its use. In the future, advanced gas sensors with integrated self-calibration are expected.

B. Comparison with IoT and Non-IoT System

Fig. 6 shows the difference in the efficiency of a car parking system, which is the IoT one and the non-IoT one, and also shows the problems. The IoT-enabled parking system is advanced in connectivity, immediate data acquisition, alert security, automation, and quick response, thus improving efficiency and the user experience. Table III details the performance

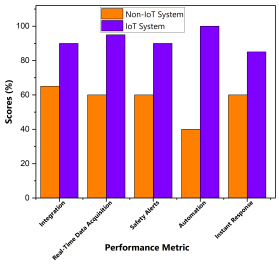


Fig. 6. Comparison the proposed system with Non-IoT System

metrics, illustrating the significant improvements achieved by the IoT system.

TABLE III
COMPARISON BETWEEN IOT AND NON-IOT SYSTEMS

Performance Metric	Non-IoT System(%)	IoT System(%)
Integration	65	90
Real-Time Data Acquisition	60	95
Safety Alerts	60	90
Automation	40	100
Instant Response	60	85

C. Real-time Analysis

This smart system's real-time analysis requires constant monitoring and reporting of various parameters that are necessary for the perfect functioning of the parking facility.

1) Mobile Application Integration: Drivers connect to the mobile application using a specific port number and IP address, enabling them to easily locate the nearest vacant parking space. The port number and IP address are displayed in Fig. 7(a). Sensor data is transmitted using the MQTT protocol to a mobile application, which drivers can use to view the status of each parking slot. Fig. 7(b) shows the application MQTT Dashboard displays vacant slots in green and occupied slots in red. Other colors indicate the inactive slot. The mobile app requires real-time communication with the MQTT broker. Until the network connection is back to normal, the system keeps the microcontroller's last known status regarding parking. With the offline mode that will be developed later, the user will be able to access some basic information on parking locally. Data sent by MQTT is encrypted with Transport Layer Security (TLS), ensuring confidentiality and integrity.

2) Entrance and Exit Gate Management: When a car comes towards the entrance, an IR-sensor detects that there is a vehicle. A buzzer goes off, notifying the car park attendant that a vehicle is wanting access to said area. Fig. Fig.7(c) illustrates the implementation of a servo motor acting as an exit gate using an IR sensor. At the entrance, there is an OLED display with how many slots exist in total and available for a car to park before it, which dynamically updates upon coming vehicles. Fig. 7(d) shows the number of all slots offered by the parking area.

3) Parking Slot Monitoring: Each parking slot is equipped with an IR sensor to detect the presence or absence of a vehicle. Fig. 7(e) shows the implementation of each slot with an IR sensor. Fig. 7(f) shows the sensors continuously send status updates to the central system and Mobile Application (MQTT Dashboard).

4) Environmental Monitoring: Real-time temperature and humidity data are generated using a DHT22 sensor, which continuously measures the environment and displays them on an OLED panel. The concentration of gas is measured by the

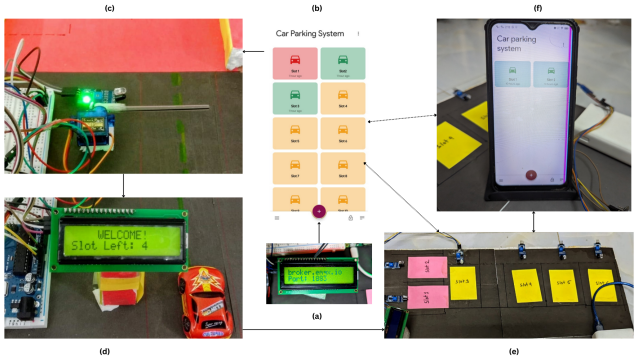


Fig. 7. (a) Displayed IP Address and Port Number, (b) Mobile Application (MQTT Dashboard), (c) Exit Gate of Parking Area, (d) Displayed Number of Free Available Slots, (e) Slots Equipped with IR Sensor, (f) Send Sensor Data to Mobile App

MQ-2 gas sensor. This device will activate in order to release part of that concentrated air through the exhaust fan in the event that the gas levels surpass a certain threshold, preserving safety.

V. Discussion, Limitations and Future Work

Our IoT-enabled car parking system integrates IR sensors, servo motors, an OLED display, and a mobile app along with environmental sensors such as DHT22 and the MQ-2 Gas Sensor. These will improve performance over traditional methods. Real-time data acquisition and smooth communication enhance the overall efficiency, while automation enhances user experience and safety. Instant parking availability updates are provided to the drivers, and an optimum environment is created and maintained. Adaptable to diverse parking designs, such as underground or multi-level facilities, the system supports hardware upgrades like ultrasonic sensors for height detection. While RFID or user-authorization systems could be considered, our design focuses on providing large-scale public parking without unnecessary complexity.

Sensor-related issues include IR sensor imprecision due to dust and light, DHT22 drift over time, and MQ-2 efficiency loss with wear. Network dependency is also one of the challenges because blackouts disrupt real-time data. Future upgrades will include self-cleaning IR sensors, advanced sensors such as LIDAR, better DHT22 calibration techniques, redundant MQ-2 gas sensors, and offline data storage in case of network outages. Predictive sensor maintenance with machine learning, enhanced security through end-to-end encryption, and integration with blockchain for the integrity of data-what is proposed for a robust, scalable, and efficient performance of the system.

VI. Conclusion

Our IoT-enabled car parking system outperforms the traditional system by integrating IR sensors, servo motors, OLED displays, DHT22 sensors, and MQ-2 gas sensors, with mobile apps for real-time communication of data and for control automation. The present challenges faced are setup costs and network dependence; the next phase will focus on cost reduction and reliability improvement. The system supports smart city initiatives by reducing emissions and optimizing urban mobility, showcasing IoT's potential to transform infrastructure with applications in traffic management, energy, and public safety, improving urban quality of life globally.

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